

Chapter 2: Communication and networking technologies

Learning objectives

By the end of this chapter you should be able to:

- show understanding of the purpose and benefits of networking devices
- show understanding of the characteristics of a LAN (local area network) and a WAN (wide area network)
- explain the client-server and peer-to-peer models of networked computers
- show understanding of thin-client and thick-client and the differences between them
- show understanding of the bus, star, mesh and hybrid topologies
- show understanding of cloud computing
- show understanding of the differences between and implications of the use of wireless and wired networks
- describe the hardware that is used to support a LAN
- describe the role and function of a router in a network
- show understanding of Ethernet and how collisions are detected and avoided
- show understanding of bit streaming
- show understanding of the differences between the World Wide Web (WWW) and the Internet
- describe the hardware that is used to support the Internet
- explain the use of IP addresses in the transmission of data over the Internet
- explain how a Uniform Resource Locator (URL) is used to locate a resource on the World Wide Web (WWW) and the role of the Domain Name Service (DNS).

2.01 The evolution of the purpose and benefits of networking

Wide area network (WAN)

During the 1970s it would be normal for a large organisation to have a computer. This computer would be a mainframe or minicomputer. The computer could have been running a time-sharing operating system with individual users accessing the computer using a terminal connected to the computer with a cable. Technology was developed that allowed computers in different organisations to be networked using what would now be described as a **wide area network (WAN)**. In a WAN, the networked computers could be thousands of kilometres apart.

The benefits of having the computers connected by a WAN were:

- a 'job' could be run on a remote computer that had the required application software
- a data archive that was stored on a remote computer could be accessed
- a message could be transmitted electronically to a user on a remote computer.

Today, a typical WAN is characterised by the following.

- It will be used by an organisation or a company to connect sites or branches.
- It will not be owned by the organisation or company.
- It will be leased from a public switched telephone network company (PSTN).
- A dedicated communication link will be provided by the PSTN.
- The transmission medium will be fibre-optic cable.
- Transmission within the WAN will be from switch to switch.
- A switch will connect the WAN to each site.
- There will not be any end-systems connected directly to the WAN.

Local area network (LAN)

In the 1980s the arrival of the microcomputer or personal computer (PC) changed computing. In an organisation, a user could have their own computer on their desk. The computer could be a stand-alone system or some organisations chose to have more than one computer connected using a **local area network (LAN)**. It was called a local area network because it typically connected PCs that were in one room or in one building or on one site.



KEY TERMS

Wide area network (WAN): a network connecting computers on different sites, possibly thousands of kilometres apart

Local area network (LAN): a network connecting computers in a single room, in a single building or on a single site

The benefits of connecting PCs in a LAN included the following.

- The expense of installing application software on each individual PC could be saved by installing the software on an application server attached to the LAN instead.
- A file server could be attached to the LAN that allowed users to store larger files and also allowed files to be shared between users.
- Instead of supplying individual printers to be connected to a user's PC, one or more printers could be attached to a print server that was connected to the LAN; these could be higher quality printers.

- Managers in organisations could use electronic mail to communicate with staff rather than sending round memos on paper.
- The 'paper-less office' became a possibility, where files were to be stored in digital form on a file server rather than as paper copies in a filing cabinet.

Today, a typical LAN is characterised by the following.

- It will be used by an organisation or a company within a site or branch.
- It will be owned by the organisation or company.
- It will be one of many individual LANS at one site.
- The transmission medium will be twisted pair cable or WiFi.
- The LAN will contain a device that allows connection to other networks.
- There will be end-systems connected which will be user systems or servers.

Discussion Point:

If a print server was attached to a network, what functionality could it provide?

Internet working

The 1990s can be said to be when the modern era of computing and network use started, with the beginning of widespread use of the Internet. The word Internet is a shortened form of the term 'internetwork', which describes a number of networks all connected together. LANs are connected to WANs which are in turn connected to the Internet to allow access to resources world-wide. The other technologies defining the modern era, namely mobile devices and wireless networking, started to become commonly used in the 2000s.

The purpose and benefits of networking have not changed but their scale and scope has increased enormously. In particular, people now have full access to networks from their personal devices.

The client-server model

The **client-server** model (or architecture) was first used in large organisations when they had installed internal networks. Typically, the organisation would have individual LANs connected via an organisation-wide WAN. An individual LAN might have had an application server attached. The organisation was likely to need a powerful central computer. The central computer could be connected to the WAN as a server. It would probably not have individual users connected to it directly. A PC, attached to a LAN, could access the server as a client.

The client-server mode of operation nowadays is different. The client is a web browser connected to the Internet. The server is a web server hosted on the Internet.

The server provides an application and the client uses the application. There are two options for how the client functions.

A **thin-client** is one which:

- chooses an application to run on the server
- sends input data to the server when requested by the application
- receives output from the application.

KEY TERMS

Client-server: an architecture where a client runs an application provided by a server on a network

Thin-client: a client that only provides input and receives output from the application

A **thick-client** is one which:

- chooses an application provided by the server
- possibly carries out some processing before running the application on the server and also after receiving output from the application
- alternatively, possibly downloads the application from the server and runs the application itself.



TIP

In thick-client mode the processing on the client can be controlled by the use of a scripting language. You do not need to know any details of this.



KEY TERM

Thick-client: a client that carries out at least some of the processing itself

The client-server approach is the choice in the following circumstances.

- The server stores a database which is accessed from the client system.
- The server stores a web application which allows the client system to find or, sometimes, supply information.
- The server stores a web application which allows the client system to carry out an e-commerce or financial transaction.

File sharing

If a user uploads files to a file server then the client-server operation can be used by another user to download these from the server.

An alternative mode of operation for sharing files is peer-to-peer networking. Instead of having one server that many clients access, a peer-to-peer network operates with each peer (networked computer) storing some of the files. Each peer can therefore act as a client and request a file from another peer or it can act as a server when another peer requests the download of a file.

The peer-to-peer model has several advantages compared to client-server file downloading:

- it avoids the possibility of congestion on the network when many clients are simultaneously attempting to download files
- parts of a file can be downloaded separately
- the parts are available from more than one host.

The client-server model has the following advantages.

- It allows an organisation to control the downloading and use of files.
- The files can be better protected from malware attacks because the files are stored on one server which will be regularly scanned using appropriate anti-virus software.

2.02 Network topologies

There are five requirements for a data communications system: a sender, a receiver, a transmission medium, a message and a protocol (see Chapter 17 for details about protocols). A transmission medium can be air (e.g. for WiFi) or cables (e.g. for Ethernet). Data can be sent through the medium in different modes: